



Working with Yocto to build Linux image

Linux host setup

Introduction

The Yocto build system requires a Linux host machine. You can either run this host as a standalone native computer or as a virtual machine

Required Packages

The Yocto Project requires several packages to be installed on the host machine.

```
$ sudo apt-get install gawk wget git-core diffstat unzip texinfo  
gcc-multilib build-essential chrpath socat libsdl1.2-dev xterm sed cvs subversion  
coreutils texi2html docbook-utils python-pysqlite2 help2man make  
gcc g++ desktop-file-utils libgl1-mesa-dev libglu1-mesa-dev  
mercurial autoconf automake groff curl lzip asciidoc u-boot-tools
```

Install the repo tool

The repo tool has been developed to make it easier to manage multiple Git repositories. Instead of downloading each repository separately the repo tool can download all with one instruction. Download and install the tool by following the instructions below.

```
$ mkdir -p ~/bin  
$ curl https://storage.googleapis.com/git-repo-downloads/repo > ~/bin/repo  
$ chmod a+x ~/bin/repo  
$ export PATH=~/bin:$PATH
```

Configure Git

If you haven't already done so. Change "Your name" to your actual name and "your e-mail" to your e-mail address.

```
$ git config --global user.email "your@e-mail"  
$ git config --global user.name "Your Name"
```

Download source branch

Introduction

All available branches are configured for Linux version: 6.6.23 and U-Boot version: 2024.04.
Yocto settings are MACHINE: imx8mm-smart DISTRO: fsl-imx-wayland

Branch folder

Create a directory for the downloaded files

```
$ mkdir branch_dir  
$ cd branch_dir
```

Initialize repo

This branch provides a base functionality and can use for SDK. It also available with Mender update support (OTA update), minimal dashboard (to configure WIFI), an Influx developed service, to connect with Rexgen (rexgend)

```
$repo init -u https://github.com/InfluxTechnology/influx-yocto-base -b influx-6.6.23 -m  
base.xml
```

All files have will be downloaded into the <branch_dir>.
Most of the files will actually be available in the sub-directory called /sources

```
$ repo sync
```

Build image

Introduction

Yocto is using the bitbake tool to generate complete Linux images/distributions, that is, all needed to boot and run a Linux system. This is typically boot loader(s), Linux kernel, and root file system with selected utilities and applications

Initialize build

Before starting the build, it must be initialized. In this step, the build directory and local configuration files are created.

```
$ source influx-setup-release.sh -b build_dir
```

Starting the build

Everything has now been set up to start the actual build. Please note that building an image can take many hours, depending on your host computer's capabilities.

```
$ bitbake influx-image-base
```

Restart the build

If you need to restart a build in a new terminal window or after a restart of the host computer, you don't need to rerun the influx-setup script. Instead, you run the setup-environment script.

```
$ source setup-environment build_dir
```

Deploying image

Introduction

Build process stores the files in <build_dir>/tmp/deploy/images/imx8mm-smart/. We have a script, that collect all needed files: sdimg, wic, dtb, bin and create an archive <build_dir>/uuu-deploy.tgz

\$./deploy-image.sh

UUU

UUU (Universal Update Utility) can be used to write images to the board. This tool is sending files and instructions over USB and the board must be set in OTG boot mode for it to work. it is publicly available on GitHub (<https://github.com/NXPmicro/mfgtools>). In archive uuu-deploy.tgz have two instances of this tool, build for Linux and Windows.

Deploy image

Copy and extract the archive uuu-deploy.tgz. Go to /uuu-deploy folder and start the deploying process.

\$ sudo ./uuu deploy-image-sdimg.uuu

For Windows OS

> uuu.exe deploy-image-sdimg.uuu